Continued Fractions

Nick Wendt

December 18, 2025

Definition

A simple continued fraction is an expresion of the form

$$a_0 + \cfrac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + ...}}}$$

Where $a_0 \in \mathbb{Z}$ and $a_1, ..., a_n \in \mathbb{N}$
We write a continued fraction compactly as $[a_0; a_1, a_2, a_3, ...]$

Simple Example

Write $\frac{31}{12}$ as a continued fraction Step 1: Find the floor of $\frac{31}{12}$

$$\left\lfloor \frac{31}{12} \right\rfloor = 2$$

This is your a_0

Step 2: Subtract your original number by the floor calculated in step 1.

$$\frac{31}{12} - 2 = \frac{31}{12} - \frac{24}{12} = \frac{7}{12}$$

This will always give you a number between 0 and 1.

Step 3: Flip the fraction

$$\frac{7}{12} \to \frac{12}{7}$$

Step 4: Repeat steps 1-3 until you get a 1 as the numerator of the fraction in step 2.

Finishing the example:

$$Floor: \left\lfloor \frac{12}{7} \right\rfloor = 1 = a_1$$
$$Subtract: \frac{12}{7} - 1 = \frac{5}{7}$$
$$Flip: \frac{5}{7} \to \frac{7}{5}$$
$$Repeat$$
$$Floor: \left\lfloor \frac{7}{5} \right\rfloor = 1 = a_2$$
$$Subtract: \frac{7}{5} - 1 = \frac{2}{5}$$
$$Flip: \frac{2}{5} \to \frac{5}{2}$$
$$Floor: \left\lfloor \frac{5}{2} \right\rfloor = 2 = a_3$$
$$Subtract: \frac{5}{2} - 2 = \frac{1}{2}$$
$$Stop$$

We stop since 1 is in the numerator of the fraction.

Now using our $a_i's$, we can write our continued fraction:

$$\frac{31}{12} = 2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2}}}}$$

Irrational Example

Recurrence Relations

The nth convergent approximation of a number is $[a_0; a_1, a_2, ..., a_n] = \frac{p_n}{q_n}$ where $gcd(p_n, q_n) = 1$
It follows that $a_0 = [a_0] = \frac{p_0}{q_0}$ so $p_0 = a_0$ and $q_0 = 1$
Then $p_1 = a_0 * a_1 + 1$ and $q_1 = a_1$

The recurrence relations are $p_n = a_n * p_{n-1} + p_{n-2}$ and $q_n = a_n * q_{n-1} + q_{n-2}$

Proof

Assume:

$$\frac{p_{n-1}}{q_{n-1}} = \frac{a_{n-1} * p_{n-2} + p_{n-3}}{a_{n-1} * q_{n-2} + q_{n-3}}$$

Then the RHS equals:

$$\frac{(a_{n-1}+\frac{1}{a_n})p_{n-2}+p_{n-3}}{(a_{n-1}+\frac{1}{a_n})q_{n-2}+q_{n-3}}$$

because $[a_0; a_1, a_2, ..., a_{n-1}, a_n] = [a_0; a_1, a_2, ..., a_{n-1} + \frac{1}{a_n}]$ by definition of continued fractions now multiply by $\frac{a_n}{a_n}$ and see the recurrence relation appear.

Lagrange's Theorem

A real number has a periodic continued fraction expansion $[a_0; a_1, ..., a_n, \overline{a_{n+1}, ..., a_{n+k}}]$ if and only if it (α) is a solution of a quadratic equation: $\alpha = r + s\sqrt{m}$ s.t. $r, s \in \mathbb{Q}$ and $m \in \mathbb{Z}_{\geq 2}$ and m is not a square.